Compute and Disk

Compute

Every project on the Supabase Platform comes with its own dedicated Postgres instance.

The following table describes the base instances, Nano (free plan) and Micro (paid plans), with additional compute instance sizes available if you need extra performance when scaling up.

Nano instances in paid plan organizations

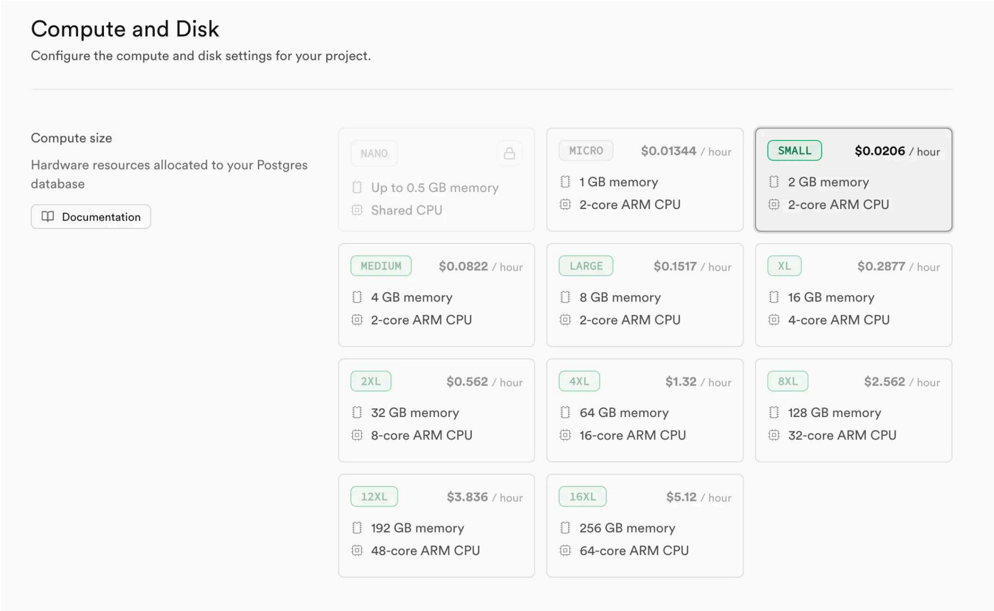
In paid organizations, Nano Compute are billed at the same price as Micro Compute. It is recommended to upgrade your Project from Nano Compute to Micro Compute when it's convenient for you. Compute sizes are not auto-upgraded because of the downtime incurred. See Supabase Pricing for more information. You cannot launch Nano instances on paid plans, only Micro and above - but you might have Nano instances after upgrading from Free Plan.

Compute Size	Hourly Price USD	Monthly Price USD	CPU	Memory	Max DB Size (Recommended)[1]
Nano[2]	$0	$0	Shared	Up to 0.5 GB	500 MB
Micro	$0.01344	~$10	2-core (shared)	1 GB	10 GB
Small	$0.0206	~$15	2-core (shared)	2 GB	50 GB

Skip to content

Compute Size	Hourly Price USD	Monthly Price USD	CPU	Memory	Max DB Size (Recommended)[1]
Medium	$0.0822	~$60	2-core (shared)	4 GB	100 GB
Large	$0.1517	~$110	2-core (dedicated)	8 GB	200 GB
XL	$0.2877	~$210	4-core (dedicated)	16 GB	500 GB
2XL	$0.562	~$410	8-core (dedicated)	32 GB	1 TB
4XL	$1.32	~$960	16-core (dedicated)	64 GB	2 TB
8XL	$2.562	~$1,870	32-core (dedicated)	128 GB	4 TB
12XL	$3.836	~$2,800	48-core (dedicated)	192 GB	6 TB
16XL	$5.12	~$3,730	64-core (dedicated)	256 GB	10 TB
>16XL	-	Contact Us	Custom	Custom	Custom

Compute sizes can be changed by first selecting your project in the dashboard here and the upgrade process will incur downtime.



We charge hourly for additional compute based on your usage. Read more about usage-based billing for compute.

Dedicated vs shared CPU

All Postgres databases on Supabase run in isolated environments. Compute instances `Nano` to `2XL` compute size have CPUs which can burst to higher performance levels for short periods of time. Instances bigger than `Large` have predictable performance levels and do not exhibit the same burst behavior.

Compute upgrades

> Compute instance changes are usually applied with less than 2 minutes of downtime, but can take longer depending on the underlying Cloud Provider.

When considering compute upgrades, assess whether your bottlenecks are hardware-constrained or software-constrained. For example, you may want to look into optimizing the number of connections or examining query performance. When you're happy with your Postgres instance's performance, then you can focus on additional compute resources. For example, you can load test your application in staging to understand your compute requirements. You can also start out on a smaller tier, create a report in the Dashboard to monitor your CPU utilization, and upgrade as needed.

Disk

Supabase databases are backed by high performance SSD disks. The *effective performance* depends on a combination of all the following factors:

- Compute size
- Provisioned Disk Throughput
- Provisioned Disk IOPS: Input/Output Operations per Second, which measures the number of read and write operations.
- Disk type: io2 or gp3
- Disk size

> The disk size and the disk type dictate the maximum IOPS and throughput that can be provisioned. The effective IOPS is the lower of the IOPS supported by the compute size or the provisioned IOPS of the disk. Similarly, the effective throughout is the lower of the throughput supported by the compute size and the provisioned throughput of the disk.

The following sections explain how these attributes affect disk performance.

Compute size

The compute size of your project affects the effective disk throughput and IOPS. The table below shows both the baseline (sustained) limits and the burst (maximum) limits for each instance size. For instance, an 8XL compute instance has a throughput of 1,188 MB/s and IOPS of 40,000.

Compute Instance	Baseline Throughput (MB/s)	Max Throughput (MB/s)	Baseline IOPS	Max IOPS
Nano (free)	5 MB/s	261 MB/s	250 IOPS	11,800 IOPS
Micro	11 MB/s	261 MB/s	500 IOPS	11,800 IOPS
Small	22 MB/s	261 MB/s	1,000 IOPS	11,800 IOPS
Medium	43 MB/s	261 MB/s	2,000 IOPS	11,800 IOPS
Large	79 MB/s	594 MB/s	3,600 IOPS	20,000 IOPS
XL	149 MB/s	594 MB/s	6,000 IOPS	20,000 IOPS
2XL	297 MB/s	594 MB/s	12,000 IOPS	20,000 IOPS
4XL	594 MB/s	594 MB/s	20,000 IOPS	20,000 IOPS
8XL	1,188 MB/s	1,188 MB/s	40,000 IOPS	40,000 IOPS
12XL	1,781 MB/s	1,781 MB/s	50,000 IOPS	50,000 IOPS
16XL	2,375 MB/s	2,375 MB/s	80,000 IOPS	80,000 IOPS
24XL	3,750 MB/s	3,750 MB/s	120,000 IOPS	120,000 IOPS
24XL - Optimized CPU	3,750 MB/s	3,750 MB/s	120,000 IOPS	120,000 IOPS
24XL - Optimized Memory	3,750 MB/s	3,750 MB/s	120,000 IOPS	120,000 IOPS

Compute Instance	Baseline Throughput (MB/s)	Max Throughput (MB/s)	Baseline IOPS	Max IOPS
24XL - High Memory	3,750 MB/s	3,750 MB/s	120,000 IOPS	120,000 IOPS
48XL	5,000 MB/s	5,000 MB/s	240,000 IOPS	240,000 IOPS
48XL - Optimized CPU	5,000 MB/s	5,000 MB/s	240,000 IOPS	240,000 IOPS
48XL - Optimized Memory	5,000 MB/s	5,000 MB/s	240,000 IOPS	240,000 IOPS
48XL - High Memory	5,000 MB/s	5,000 MB/s	240,000 IOPS	240,000 IOPS

Smaller compute instances like Nano, Micro, Small, and Medium can burst above baseline for short periods of time. Once burst capacity is exhausted, performance returns to baseline. If you need consistent disk performance, consider upgrading your compute size.

Larger compute instances (4XL and above) are designed for sustained, high performance with specific IOPS and throughput limits which you can configure. If you hit your IOPS or throughput limit, throttling will occur.

Choosing the right compute instance for consistent disk performance

If you need consistent disk performance, choose the 4XL or larger compute instance. If you're unsure of how much throughput or IOPS your application requires, you can load test your project and inspect these metrics in the Dashboard. If the `Disk IO % consumed` stat is more than 1%, it indicates that your workload has exceeded the baseline IO throughput during the day. If this metric goes to 100%, the workload has used up all available disk IO budget. Projects that use any disk IO budget are good candidates for upgrading to a larger compute instance with higher throughput.

Provisioned disk throughput and IOPS

Skip to content

The default disk type is gp3, which comes with a baseline throughput of 125 MB/s and a default IOPS of 3,000. You can provision additional IOPS and throughput from the Infrastructure settings page, but keep in mind that the effective IOPS and throughput will be limited by the compute instance size. This requires Large compute size or above.

> ⚠ Be aware that increasing IOPS or throughput incurs additional charges.

Disk types

When selecting your disk, it's essential to focus on the performance needs of your workload. Here's a comparison of our available disk types:

	General Purpose SSD (gp3)	High Performance SSD (io2)
Use Case	General workloads, development environments, small to medium databases	High-performance needs, large-scale databases, mission-critical applications
Max Disk Size	16 TB	60 TB
Max IOPS	16,000 IOPS (at 32 GB disk size)	80,000 IOPS (at 80 GB disk size)
Throughput	125 MB/s (default) to 1,000 MB/s (maximum)	Automatically scales with IOPS
Best For	Great value for most use cases	Low latency and very high IOPS requirements
Pricing	Disk: 8 GB included, then $0.125 per GB IOPS: 3,000 included, then $0.024 per IOPS Throughput: 125 MB/s included, then $0.95 per MB/s	Disk: $0.195 per GB IOPS: $0.119 per IOPS Throughput: Scales with IOPS at no additional cost

For general, day-to-day operations, gp3 should be more than enough. If you need high throughput and IOPS for critical systems, io2 will provide the performance required.

Skip to content

Compute instance size changes will not change your selected disk type or disk size, but your IO limits may change according to what your selected compute instance size supports.

Disk size

General Purpose (gp3) disks come with a baseline of 3,000 IOPS and 125 MB/s. You can provision additional 500 IOPS for every GB of disk size and additional 0.25 MB/s throughput per provisioned IOPS.

High Performance (io2) disks can be provisioned with 1,000 IOPS per GB of disk size.

Limits and constraints

Postgres replication slots, WAL senders, and connections

Replication Slots and WAL Senders are used to enable Postgres Replication. Each compute instance also has limits on the maximum number of database connections and connection pooler clients it can handle.

The maximum number of replication slots, WAL senders, database connections, and pooler clients depends on your compute instance size, as follows:

Compute instance	Max Replication Slots	Max WAL Senders	Database Max Connections[3]	Connection Pooler Max Clients
Nano (free)	5	5	60	200
Micro	5	5	60	200
Small	5	5	90	400
Medium	5	5	120	600
Large	8	8	160	800

Compute instance	Max Replication Slots	Max WAL Senders	Database Max Connections[3]	Connection Pooler Max Clients
XL	24	24	240	1,000
2XL	80	80	380	1,500
4XL	80	80	480	3,000
8XL	80	80	490	6,000
12XL	80	80	500	9,000
16XL	80	80	500	12,000

> As mentioned in the Postgres documentation, setting `max_replication_slots` to a lower value than the current number of replication slots will prevent the server from starting. If you are downgrading your compute instance, ensure that you are using fewer slots than the maximum number of replication slots available for the new compute instance.

Constraints

You can modify disk attributes up to **four times** within a rolling 24-hour window. A new modification can be initiated as soon as the previous one completes. If you reach this limit, you will encounter throttling and must wait for the rolling 24-hour window to permit further adjustments.

You can increase disk size but cannot decrease it.

1. Database size for each compute instance is the default recommendation but the actual performance of your database has many contributing factors, including resources available to it and the size of the data contained within it. See the shared responsibility model for more information. ↩

2. Compute resources on the Free plan are subject to change. ↩

Skip to content

3 Database max connections are recommended values and can be customized via `max_connections` depending on your use case. Be aware of these considerations before modifying. ↩

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